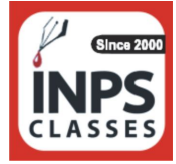




QUADRATIC EQUATION

RELATION BETWEEN ROOTS AND COEFFICIENTS

LECTURE: 03 [ADVANCE]



- 01.** Let $P_n = \alpha^n + \beta^n, n \in \mathbb{N}$. If $P_{10} = 123, P_9 = 76, P_8 = 47$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is:
[JEE MAIN 2025]
(a) $x^2 + x - 1 = 0$ (b) $x^2 + x + 1 = 0$
(c) $x^2 - x + 1 = 0$ (d) $x^2 - x - 1 = 0$
- 02.** If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$ is:
[JEE MAIN 2024]
(a) $x^2 + 8x + 12 = 0$ (b) $2x^2 + 11x + 12 = 0$
(c) $x^2 + 10x + 16 = 0$ (d) $4x^2 + 14x + 12 = 0$
- 03.** Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then, $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation [JEE Mains 2023]
(a) $7x^2 - 245x + 250 = 0$
(b) $7x^2 + 245x - 250 = 0$
(c) $49x^2 - 245x + 250 = 0$
(d) $49x^2 + 245x + 250 = 0$
- 04.** If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to
[JEE Mains 2022]
(a) 18 (b) 24
(c) 36 (d) 96
- 05.** Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to
[JEE Mains 2022]
(a) 37 (b) 58
(c) 68 (d) 92
- 06.** Let $\lambda \neq 0$ be in $\mathbb{R}$. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to [JEE Mains 2021]
(a) 18 (b) 27
(c) 9 (d) 36
- 07.** $\operatorname{cosec} 18^\circ$ is a root of the equation [JEE Mains 2021]
(a) $x^2 + 2x - 4 = 0$ (b) $4x^2 + 2x - 1 = 0$
(c) $x^2 - 2x - 4 = 0$ (d) $x^2 - 2x + 4 = 0$
- 08.** If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to
[JEE Mains 2021]
(a) 52×3^{24} (b) 56×3^{25}
(c) 28×3^{25} (d) 56×3^{24}
- 09.** If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to
[JEE Mains 2020]
(a) $\frac{9}{4}(9 + q^2)$ (b) $\frac{9}{4}(9 - q^2)$
(c) $\frac{9}{4}(9 + p^2)$ (d) $\frac{9}{4}(9 - p^2)$
- 10.** If α and β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1 - \alpha^2} + \frac{\beta}{1 - \beta^2}$ is equal to [JEE Mains 2020]
(a) 27/32 (b) 1/24
(c) 3/8 (d) 27/16

Answer Key

01. (a) 02. (a) 03. (c) 04. (b) 05. (b)
06. (a) 07. (c) 08. (a) 09. (d) 10. (d)